DRUGS · REFLUX

Alginates

Alginates react with gastric acid to form a viscous gel raft that floats on the stomach contents, reducing reflux of both acid and non-acid material, including pepsin, into the oesophagus and pharynx. Because pepsin and non-acid reflux are thought to matter in laryngopharyngeal reflux, an alginate after meals and at bedtime is a common first-line choice in UK ENT practice, alone or with a PPI.

1 CHAPTER ONE At a glance

The drugs on one page.

1.1 Summary

ALGINATES

	SUMMARY
Agents	Sodium alginate with bicarbonate and calcium carbonate (for example Gaviscon Advance, Gaviscon Double Action)
How they work	Form a gel raft above the gastric contents that acts as a physical barrier; the raft is itself refluxed preferentially
ENT uses	Suspected laryngopharyngeal reflux; reflux symptoms in pregnancy; adjunct to PPIs
Adult dose	10 to 20 ml (or 1 to 2 tablets chewed) after meals and at bedtime, up to four times daily
Child dose	Infant preparations are different products; do not use adult products in infants
Avoid	Severe renal impairment; caution with sodium restriction (heart failure, hypertension)

2 CHAPTER TWO Using them in ENT

Practice.

2.1 Laryngopharyngeal reflux

Lifestyle measures (avoid late meals, alcohol, smoking, carbonated drinks and trigger foods; lose weight; raise the head of the bed) and an alginate at bedtime are the first step; small trials suggest benefit, though the evidence base is limited. The bedtime dose protects during sleep, when laryngopharyngeal exposure is greatest.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Generally minimal: bloating, constipation or diarrhoea; sodium and calcium loads.

3.2 Interactions

Antacid components reduce absorption of some drugs (tetracyclines, quinolones, iron, levothyroxine, azoles): separate doses by about 2 hours.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

> WHAT THE EXAMINER IS LOOKING FOR

- Mechanism (raft) and why it may suit non-acid reflux.
- The bedtime dose.

4.2 Common pitfalls

COMMON PITFALLS

- Taking an alginate at the same time as levothyroxine or a quinolone.

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